GOVERNMENT OF WESTERN AUSTRALIA

# GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

*229748.1.pdf* *Geochronology Record 1913*

# 229748.1: tonalitic gneiss, New Holland Mine

***(Yilgarn Craton granites; Youanmi Terrane, Yilgarn Craton)***

**SS Romano, Y Lu, IOH Fielding, MTD Wingate and I Zibra**

## Location and sampling

SIR SAMUEL SG51-13, DEPOT SPRINGS 2942 Zone 51, MGA 250032E 6900690N
Warox Site IXZDPS000935 Sampled on 8 October 2017

This sample was collected from low outcrop on Pinnacles Station, about 5.6 km north-northeast of Five Mile Well, 3.5 km west of the New Holland mine open pits, and 2.8 km due south of the Sandstone – Agnew Road.

## Geological context

The unit sampled is a tonalitic gneiss (Fig. 1) in the hanging wall of the c. 2660 Ma Waroonga Shear Zone (WSZ), along the eastern margin of the Youanmi Terrane (Zibra et al., 2017). The Youanmi Terrane is composed of Meso- to Neoarchean granite-greenstone lithologies that range in age from c. 3.3 Ga to c. 2.72 Ga and includes gneissic-granitic rocks intruded c. 3.0 Ga to c. 2.6 Ga. The WSZ is a west-dipping, low-angle, listric structure that postdates regional east-dipping fabrics (Zibra et al., 2017). A c. 2660 Ma syntectonic monzogranite gneiss sampled about 0.7 km to the north-northeast only yielded xenocrystic zircons dated at 2849–2781 Ma with two main inherited components of 2820 ± 3 and 2802 ± 6 Ma (GSWA 212791; Wingate et al. 2021).

**Figure 1. Hand specimen image of sample 229748: tonalitic gneiss, New Holland Mine**

## Petrographic description

The sample is a weakly foliated, fine-grained tonalitic gneiss, consisting of about 55% plagioclase, 35% quartz, 6–7% biotite, 3–4% muscovite and accessory zircon. Plagioclase occurs xenoblastic, slightly elongate grains up to 3 mm in size with sharp, finely serrated boundaries with other plagioclase grains and smoothly lobate boundaries with quartz grains; twining is rare and where it occurs is very faint and finely spaced. Quartz occurs as xenoblastic, slightly elongate grains <1.5 mm in size between plagioclase grains, or as very small (<0.2 mm in diameter) round inclusions within plagioclase. Brown biotite occurs as sub-idioblastic grains to 1 mm in length. Muscovite forms idioblastic and sub-idioblastic laths to 1 mm in length, although most are <0.5 mm. Individual laths and small folia of both micas define the wavy tectonic foliation, and parallel the slightly elongate plagioclase and quartz. The only observed accessory mineral are rare zircons to 0.05 mm slender prismatic grains. The sample has been deformed and undergone grain boundary adjustment at relatively high grade to form a low interfacial free energy, semi-polygonal aggregate.

## Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, mainly subhedral to euhedral, and slightly rounded. The crystals are up to 500 µm long, and mostly elongate, with aspect ratios up to 7:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and many crystals contain high uranium, metamict zones. A CL image of representative zircons is shown in Figure 2.

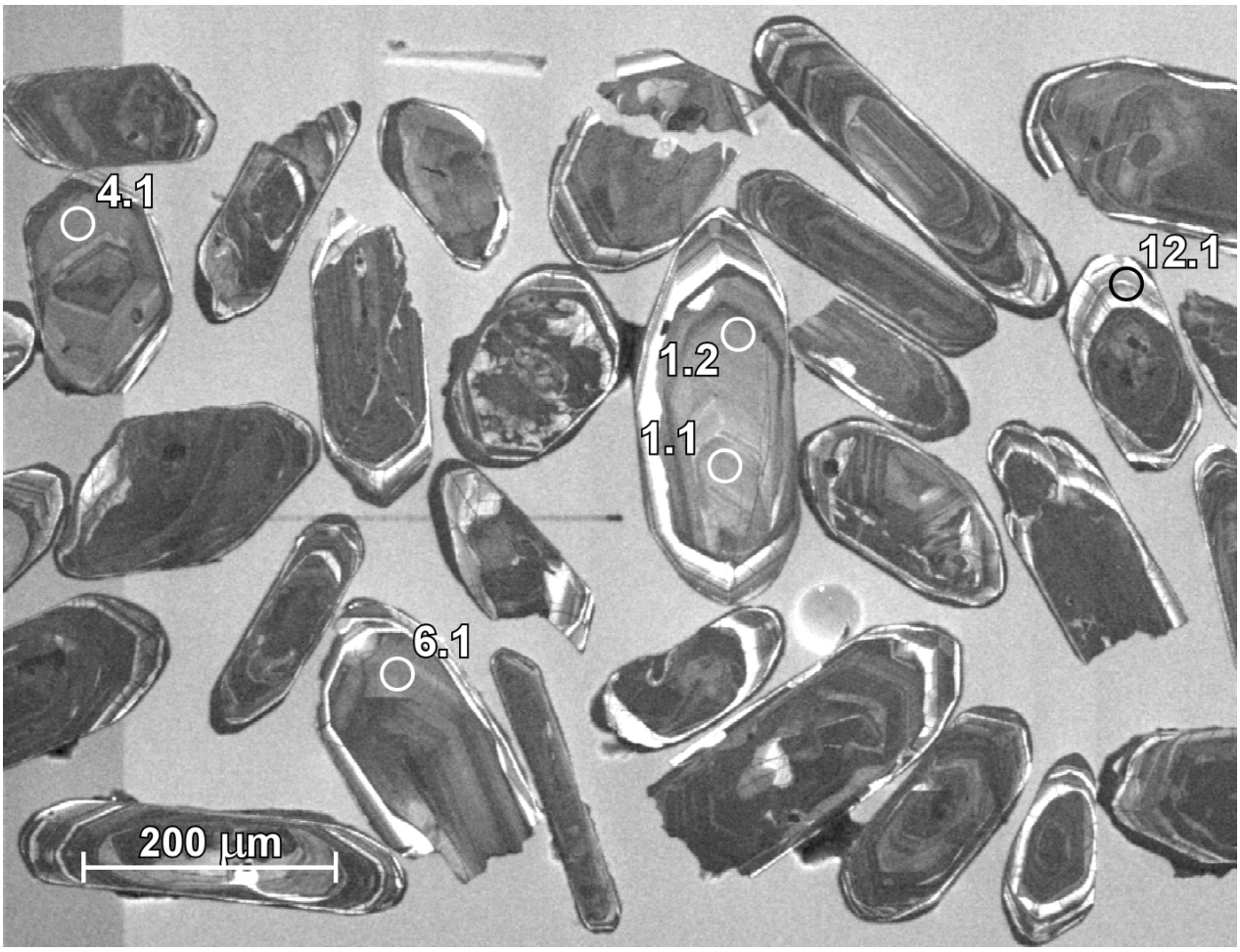


**Figure 2. Cathodoluminescence image of representative zircons from sample 229748: tonalitic gneiss, New Holland mine. Numbered circles indicate the approximate locations of analysis sites**

## Analytical details

This sample was analysed on 21–22 October 2021, using SHRIMP-B. Fifteen analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.80% (1σ) and a $^{238}U/^{206}Pb^*$calibration uncertainty of 0.32% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

## Results

Seventeen analyses were obtained from 15 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3).

## Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Six analyses are >6% discordant. The dates obtained from these six analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 11 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises nine analyses of eight zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2821 ± 10 Ma (MSWD = 1.4).

Group P comprises two analyses of two zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2782 ± 13 and 2781 ± 12 Ma (1 σ).

The date of 2821 ± 10 Ma for the nine analyses in Group I is interpreted as the igneous crystallization age of the tonalitic gneiss. The dates of 2782 ± 13 and 2781 ± 12 Ma (1 σ) for the two analyses in Group P are younger than the second analysis from the same grain and are interpreted to reflect loss of radiogenic Pb.

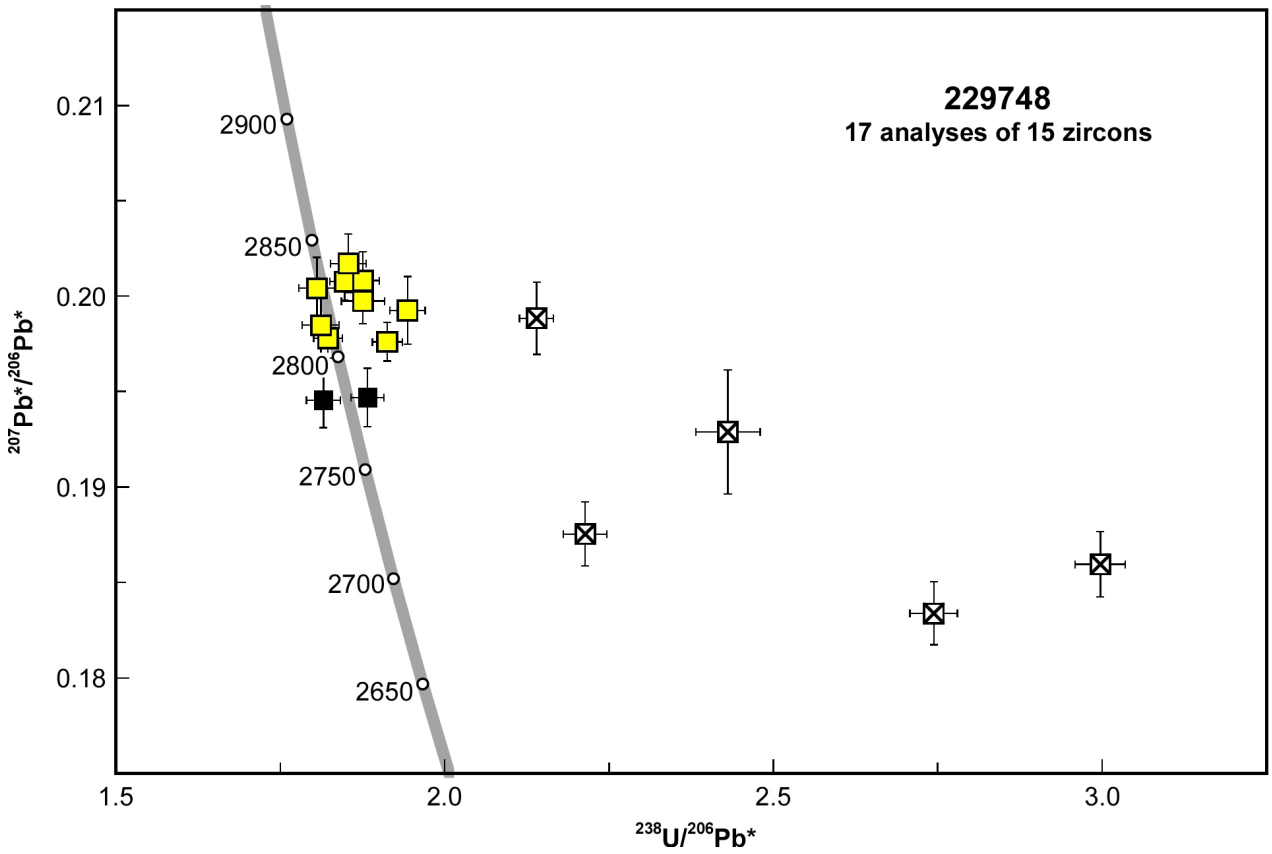


**Figure 3.** **U–Pb analytical data for zircons from sample 229748: tonalitic gneiss, New Holland mine. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >6%). One strongly discordant analysis in Group D is not shown**

## Links

Table 1. Ion microprobe analytical results for zircons from sample 229748: tonalitic gneiss, New Holland Mine: 229748.1.Table1.xls
SHRIMP spot images: 229748_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

## References


Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Wingate, MTD, Lu, Y, Fielding, IOH and Zibra, I 2021, 212791: monzogranite gneiss, New Holland mine; Geochronology Record 1824: Geological Survey of Western Australia, <www.dmirs.wa.gov.au/geochron>. View Reference online

Zibra, I, Korhonen, FJ, Peternell, M, Weinberg, RF, Romano, SS, Braga, R, De Paoli, MC and Roberts, M 2017, On thrusting, regional unconformities and exhumation of high-grade greenstones in Neoarchean orogens: the case of the Waroonga Shear Zone, Yilgarn Craton: Tectonophysics, v. v. 712–713, p. 362–395, doi:10.1016/j.tecto.2017.05.017.


### *Recommended reference for this publication*


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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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